Axioms, Conditioning & Bayes — Complete Formula Reference

Every identity from Ch. 1.2, zero derivations, plus the extra machinery your hard-problem set actually needed.

1. Kolmogorov Axioms

- **Axiom 1 (non-neg.):** $P(A) \geq 0$ for every event A
- **Axiom 2 (normalization):** $P(\Omega) = 1$
- **Axiom 3 (countable additivity):** A_i disjoint $\Rightarrow P(\bigcup A_i) = \sum P(A_i)$
- **Monotonicity:** $A \subseteq B \Rightarrow P(A) \leq P(B)$
- **Complement:** $P(A^c) = 1 - P(A)$
- **2-set inclusion–exclusion:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

trap: a probability assignment can satisfy axiom 2 (sums to 1) while still violating axiom 1 (a negative entry), and vice versa — check both, independently.

2. Union Bounds & Bonferroni (incl. chained)

- **Boole's inequality:** $P(\bigcup A_i) \leq \sum P(A_i)$
- **2-event reverse bound (lower bound on $\cap$):** $P(A \cap B) \geq P(A) + P(B) - 1$
- **Chained (3-event):** $P(A \cap B \cap C) \geq P(A)+P(B)+P(C) - 2$ (clip to 0 if negative)
- **Bonferroni sandwich (n events):** truncate PIE after S_1 (odd #terms) $\rightarrow$ upper bound; after $S_1 - S_2$ (even #terms) $\rightarrow$ lower bound; alternates thereafter

trap: Boole's bound is only informative when $n \times p \ll 1$. If $n \times p \gtrsim 1$ (e.g. the classic 52-card matching problem, where it equals exactly 1), the bound is vacuous — fall back to the exact PIE/complement value.

3. Independence: Pairwise vs. Mutual

- **Pairwise:** $P(A_i \cap A_j) = P(A_i)P(A_j)$ for every pair $i \neq j$
- **Mutual (n events):** pairwise *and* every higher-order subset product holds: $P\big(\bigcap_{i \in S} A_i\big) = \prod_{i \in S} P(A_i)$ for *all* S, $|S| \geq 2$
- **Classic knife-edge example:** A, B=coin faces, C=match — pairwise independent *only* at $p = \frac{1}{2}$; any bias $p \neq \frac{1}{2}$ breaks it

trap: if any one variable is a deterministic function of the others (e.g. $Z = X \oplus Y$), mutual independence is impossible by construction, *no matter how clean the pairwise numbers look. Checking all pairs is never a substitute for checking the full joint.*

4. Conditional Probability & Chain Rule

- **Definition:** $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, requires $P(B) > 0$
- **Chain rule (n events):** $P(A_1 \cap \cdots \cap A_n) = P(A_1)\,P(A_2 \mid A_1)\,P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$

trap: $P(A \mid B) \neq P(B \mid A)$ in general ("confusion of the inverse"). Whenever a question seems to swap these, that's the cue to use Bayes' theorem, not the plain definition.

5. Law of Total Probability

- **Partition $\{B_i\}$:** $P(A) = \sum_i P(A \mid B_i)\,P(B_i)$
- **Hidden-parameter / repeated-trial version:** if the *same* hidden parameter governs multiple trials, condition on it *first*, then combine — never just multiply/square the marginal
- **Two-stage posterior-predictive:** (1) Bayes-update the hidden-parameter posterior on observed data; (2) LTP-predict the *next* event using that *posterior*, not the original prior

trap: repeated trials under a shared hidden parameter are correlated unconditionally, *even though i.i.d.* given *the parameter — this is exactly how regime uncertainty induces correlation between asset returns.*

6. Bayes' Theorem — Standard & Odds Form

- **Standard form:** $P(A \mid B) = \dfrac{P(B \mid A)P(A)}{P(B)} = \dfrac{P(B \mid A)P(A)}{\sum_i P(B \mid A_i)P(A_i)}$
- **Odds form:** Posterior Odds = Prior Odds $\times$ Bayes Factor
- **Bayes factor (positive result):** $\dfrac{P(\text{evidence} \mid H)}{P(\text{evidence} \mid H^c)} = \dfrac{\text{sensitivity}}{\text{false-positive rate}}$
- **Bayes factor (negative result):** $\dfrac{1 - \text{sensitivity}}{1 - \text{false-positive rate}}$ — *not* the same fraction as above
- **Sequential update (cond. independent evidence):** multiply successive Bayes factors
- **Convert:** $\text{odds} = \dfrac{P}{1-P}$; $\quad P = \dfrac{\text{odds}}{1 + \text{odds}}$

trap: if two pieces of evidence share a hidden confounder, $P(E_1 \cap E_2 \mid H^c) \neq P(E_1 \mid H^c)P(E_2 \mid H^c)$ — you can no longer chain independent Bayes factors and must use the joint likelihood directly (correlated false positives inflate false confidence badly).

7. Monty Hall Family

- **Classic (3 doors, knowledgeable host):** $P(\text{win} \mid \text{switch}) = \frac{2}{3}$
- **N-door generalization:** $P(\text{win} \mid \text{switch}) = \dfrac{N-1}{N}$; $P(\text{win} \mid \text{stay}) = \dfrac{1}{N}$
- **Monty Fall (host opens randomly, happens to reveal a goat):** 50–50, *no* switching advantage
- **Biased host (prefers a specific door w.p. q when free to choose):** redo full Bayes conditional on *which specific door* opened; unconditional switch-advantage is still $\frac{N-1}{N}$ overall, but shifts between observations

trap: the mechanism generating the evidence (does the host know? does the host have a known bias? did the reveal happen by luck?) changes the answer completely, even when the visible outcome looks identical.

8. Bertrand's Box & Boy-or-Girl Family

- **Bertrand's Box (uniform prior):** $P(\text{other coin gold} \mid \text{drew gold}) = \frac{2}{3}$
- **Non-uniform box priors:** redo Bayes with the given prior weights — never reuse $\frac{2}{3}$ from memory
- **Boy-or-Girl, "at least one boy":** $P(\text{both boys}) = \frac{1}{3}$
- **Boy-or-Girl, one *specific* child identified** (elder / randomly met): $P(\text{both boys}) = \frac{1}{2}$
- **"Tuesday Boy" escalation:** adding an ever-rarer identifying detail drifts the answer from $\frac{1}{3}$ toward $\frac{1}{2}$ (e.g. $\frac{13}{27}$ for a 1-in-7 detail)

trap: count outcomes (coins/children), not boxes/categories *— and always ask exactly what random experiment generated your piece of information before conditioning.*

9. Base-Rate Neglect / Natural Frequencies

- **Natural-frequency method:** over a round population N: true-condition count $= N \times$base rate; $TP =$ that count $\times$ sensitivity; $FP = (N - \text{true count}) \times \text{FPR}$
- **Result:** $P(\text{true} \mid \text{flagged}) = \dfrac{TP}{TP + FP}$

trap: even 99%/99% accuracy can yield < 1% true-positive flags when the base rate is extreme enough — always sanity-check with whole numbers before trusting a percentage.

Quick Decision Guide — "Which Tool Do I Need?"

- Assignment of numbers to outcomes looks like a probability → check **both** Kolmogorov axioms separately, not just one.
- "At least one of / union of" several events, want a fast bound → **Boole/Bonferroni**; check $n \times p$ against 1 before trusting the bound.
- Sum of probabilities exceeds 1 → instantly gives a nontrivial **lower bound** on the intersection (reverse Bonferroni).
- Checked all pairs and they're independent → that is *not* enough for $n \geq 3$; check whether one variable determines another.
- "Given B, what's the chance of A" → plain **conditional probability**; if the question secretly wants $P(B \mid A)$ instead, that's confusion of the inverse.
- A probability depends on which of several scenarios holds → **Law of Total Probability**; if the scenario is a *hidden, repeated* parameter, update first, predict second.
- "Given evidence, what's the probability of the hypothesis" → **Bayes' theorem**; chaining multiple pieces of evidence → use **odds form**; evidence came back *negative* → use the miss-rate Bayes factor, not sensitivity.
- Two pieces of evidence might share a common cause → check for **correlated likelihoods** before chaining Bayes factors.
- Doors/boxes/reveals with a host or informant who deliberately avoided something → **Monty Hall**-style Bayes; if the reveal was random/lucky instead, the answer changes completely.
- "Naive answer looks obviously right" on a probability riddle → that's the signal you're in **Section III** (classical trap) territory — re-derive from the definition, don't trust intuition.

Numbers Worth Memorizing Cold

- $1/e \approx 0.3679$; $e \approx 2.71828$
- Monty Hall (3-door): switch wins $2/3 \approx 0.667$
- Bertrand's Box (uniform prior): $2/3 \approx 0.667$
- Boy-or-Girl: "at least one boy" → $1/3$; specific child → $1/2$
- "Tuesday Boy" extension: $13/27 \approx 0.481$
- 52-card matching problem: Boole's bound $= 1$ (exact, vacuous); true value $\approx 1 - 1/e \approx 0.632$
- Odds↔prob quick table: odds 1:99 → $P \approx 0.0099$; odds 1:1 → $P = 0.5$; odds 9:1 → $P = 0.9$